Solutions to Linear Algebra Problems

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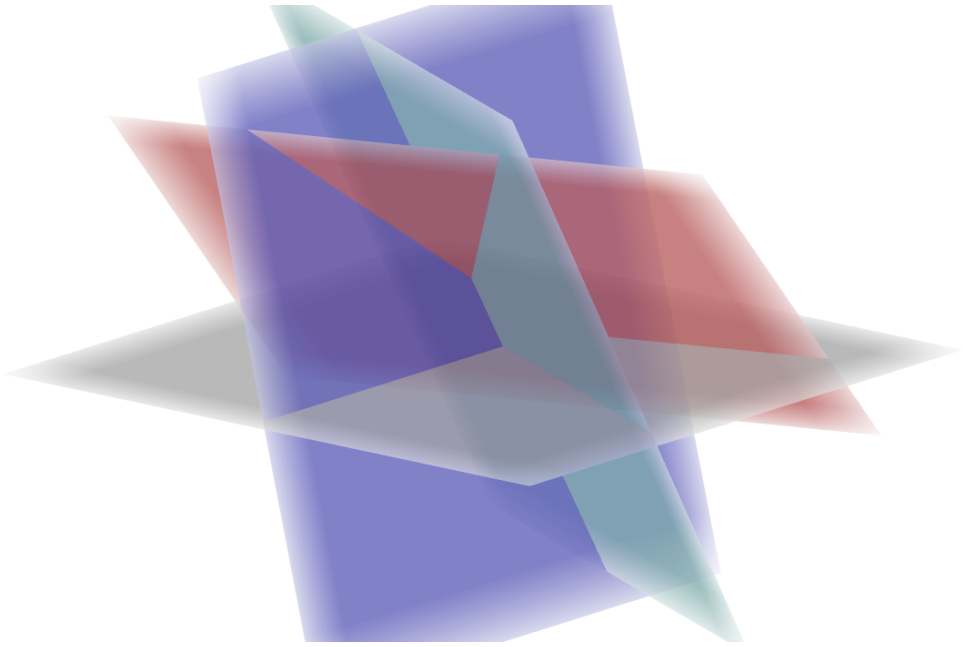
Abstract

This note collects my solutions to selected problems in Linear Algebra. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- Linear Algebra Done Right 4e, Sheldon Axler



Problem 1.

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Source: Linear Algebra Done Right - Ex.12 p.25

Solution:

Let V_1, V_2 be two subspaces of V.

($\Rightarrow$): Suppose that $V_1 \cup V_2$ is a subspace of V. Suppose for contradiction that one of the subspaces V_1 or V_2 is not contained in the other. We can choose u, w such that $u \in V_1, u \notin V_2$ and $w \in V_2, w \notin V_1$.

Since $V_1 \cup V_2$ is a subspace, we have $u + w \in V_1 \cup V_2$. This means $u + w$ must be in V_1 or V_2.

If $u + w \in V_1$, since V_1 is a subspace and $-u \in V_1$, then $w = (u + w) - u \in V_1$, which is a contradiction. Similarly, if $u+w \in V_2$, since V_2 is a subspace and $-w \in V_2$, then $u = (u+w)-w \in V_2$, which is also a contradiction.

Therefore, $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.

($\Leftarrow$): If $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$, then $V_1 \cup V_2 = V_2$ or $V_1 \cup V_2 = V_1$, which is a subspace.